

3. The water that is used in the home goes through a number of treatment steps.

- (a) The steps that water undergoes before it passes into the home are given below. They are **not** in the correct order.

Step	Process	Description
A	Distribution	Water is transported through pipes underground
B	Filtration	Water is filtered through layers of sand and gravel removing small insoluble particles
C	Sedimentation	Large insoluble particles settle out by gravity
D	Chlorination	Water is disinfected to kill any remaining bacteria to make it safe to drink
E	Use	Water is used in the home
F	Collection	Water is collected and stored in reservoirs.

Arrange the steps in the **correct** order by completing the boxes below.
The first and last boxes have been completed for you.

[3]



- (b) Pollutants can be found in drinking water making it unsafe to drink. Three students, Andrew, Robert and Malcolm investigated the solubility of a number of different salts at different temperatures.

Describe a method they could follow to find the solubility of a salt at different temperatures. [6 QER]

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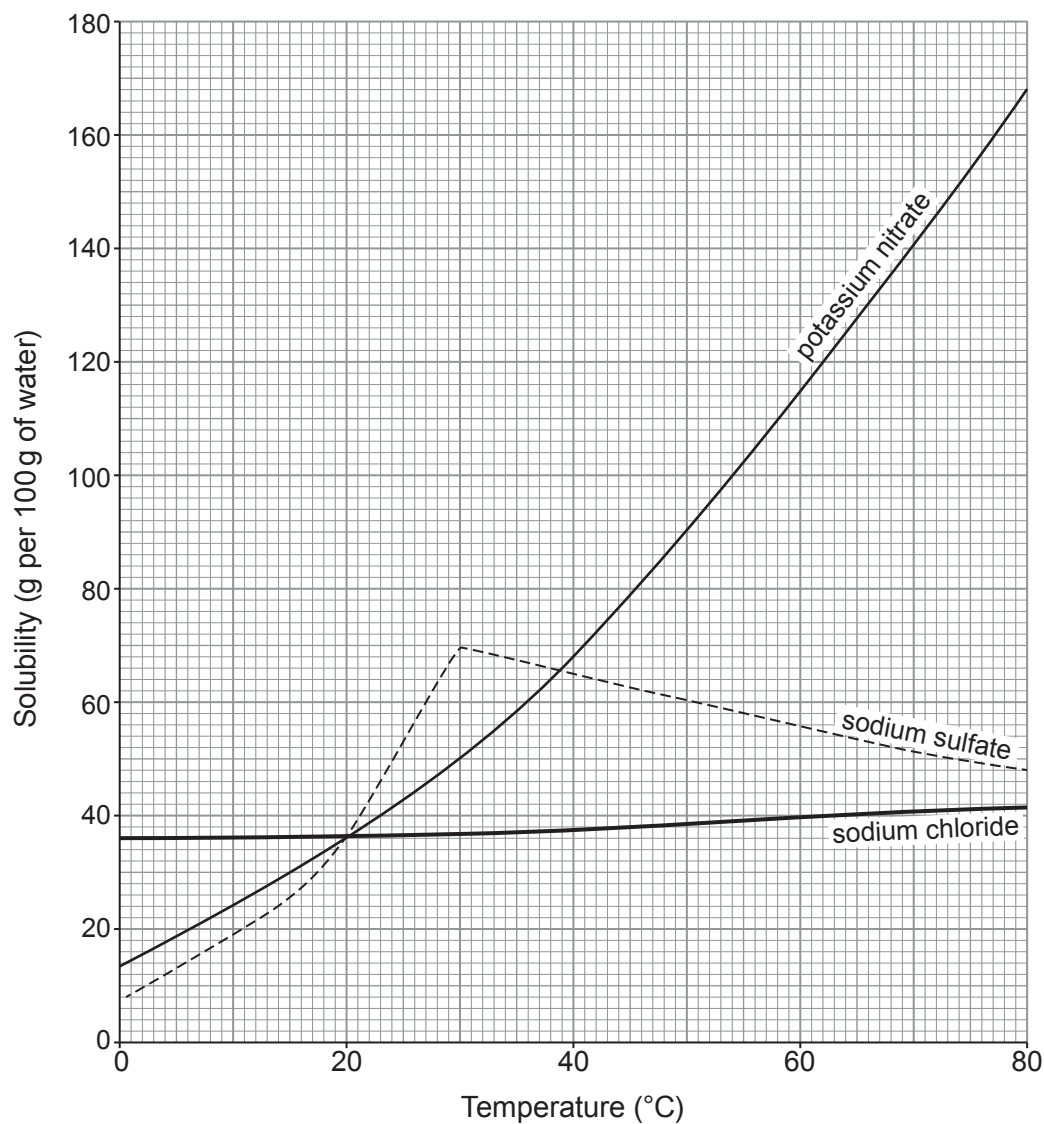
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- (c) The results of an investigation into the solubility of three salts are shown in the graph below.



- (i) State which salt is the most soluble at 30 °C.

[1]

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- (ii) Calculate the difference in solubility between potassium nitrate and sodium chloride at 60 °C.

[1]

Answer = g/100g of water

- (iii) Compare the solubilities of potassium nitrate and sodium sulfate between 30 °C and 60 °C. [3]

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- (d) Sam, a school technician, found a salt in an unlabelled container. She followed the same method as the students and her results are shown below.

Temperature (°C)	Solubility of unknown salt (g per 100 g of water)
10	24.0
30	52.0
50	120.0

Sam compared her results with the graph and identified the salt as potassium nitrate. Explain why Sam made this conclusion and whether she was correct to do so. [2]

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